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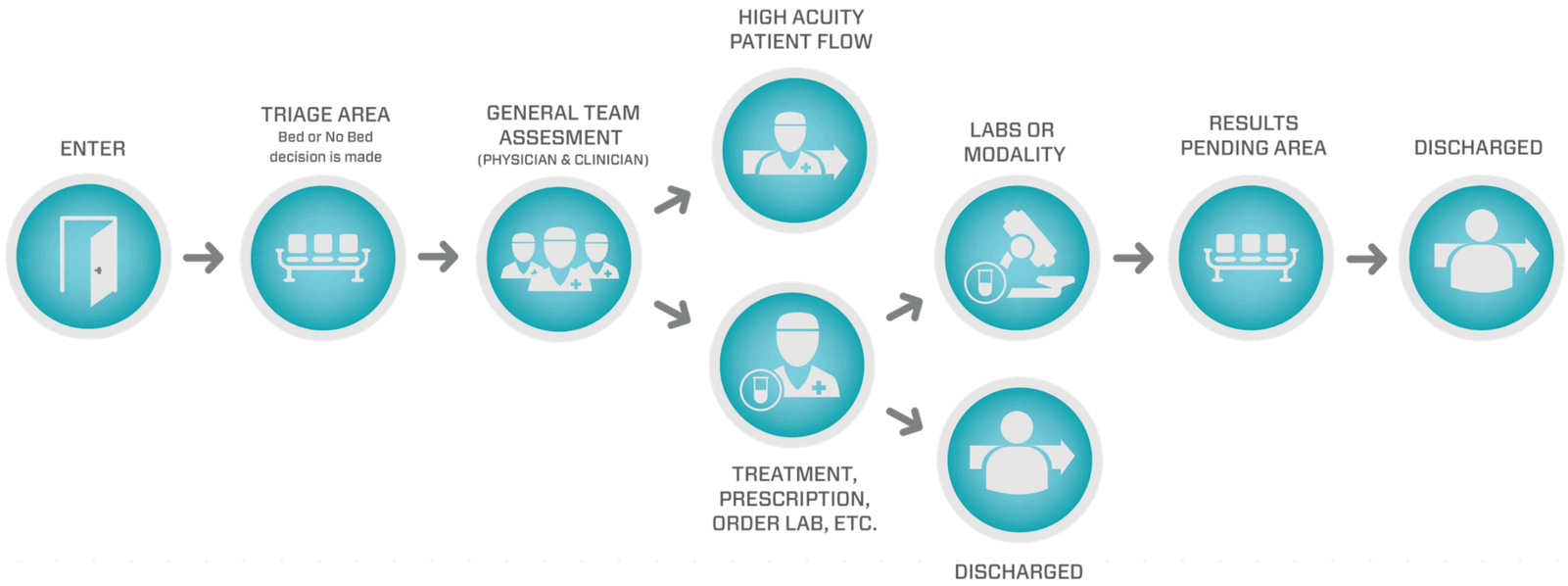
ASA DataFest

WHY PATIENTS DON'T COME BACK: AN ANALYSIS OF FOLLOW-UP CARE GAPS

Presented by: Trix

29 March, 2026

PATIENT JOURNEY FOR EMERGENCY DEPARTMENT VISIT





“Discharge”

≠

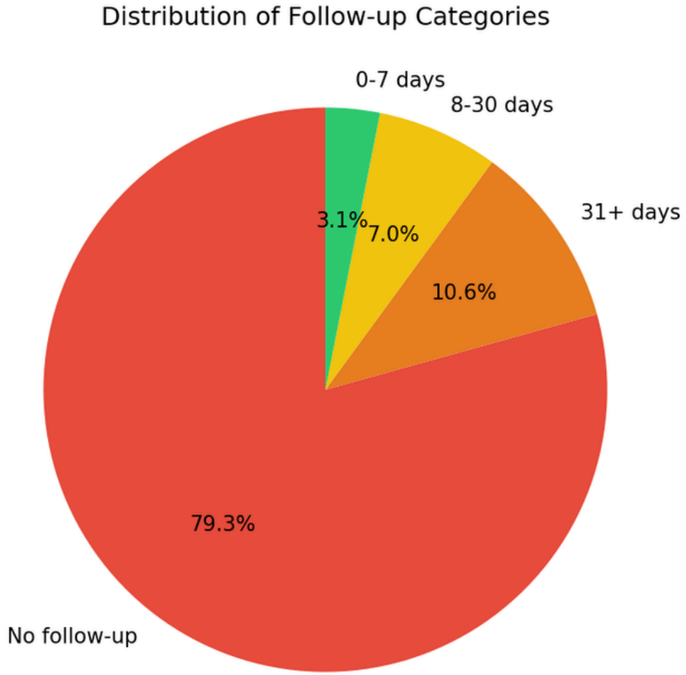


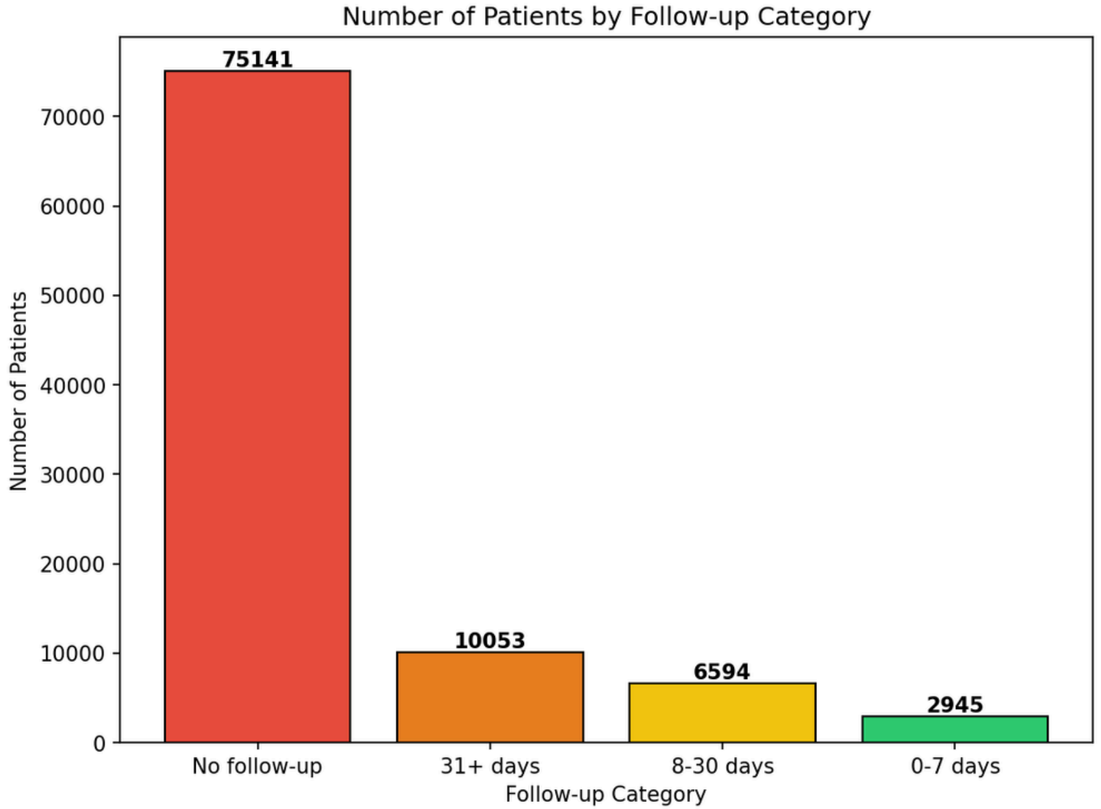
“Fully Recovered”

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Follow-up Care After Emergency Department Visits





79.3% of patients never returned after their ED visit

OBJECTIVE

To uncover why patients delay or miss follow-up care after Emergency Department visits, for targeted healthcare interventions.

Our Focus:

What is the follow-up pattern after ED visits?

→ Distribution of delay times

→ Percentage of delayed vs no follow-up patients

What factors might explain delays?

→ Social barriers (transport, financial)

→ Patient engagement

→ Demographics

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DATA SOURCES & TOOLS

DATA SOURCES USED

- **Encounters** dataset
- **Diagnosis** dataset
- **Patients** dataset
- **Social Determinants** of Health (SDOH) dataset

TOOLS USED

- **Python (Pandas, NumPy)** → data cleaning & processing
- **Jupyter Notebook** → analysis workflow
- **Scikit-learn** → Random Forest model
- **Matplotlib / Seaborn** → visualization

METHODOLOGY

Data Cleaning and Processing

01

Merged multiple datasets (Encounters, Diagnosis, Patients, SDOH)

02

Handled missing and inconsistent values

03

Filtered relevant cases: ED visits and admissions only

04

Created Key Variable (Feature Engineering)

- DelayDays (time to follow-up)
- Social barriers (transport, financial)
- Engagement (MyChart status)

APPROACH

- Link each ED visit → next visit (same/related diagnosis)
- Categorize follow-up:
  - Fast (≤ 7 days)
  - Moderate (8–30 days)
  - Delayed (> 30 days)
  - No follow-up

ANALYSIS & MODELING

01

Explore delay patterns across:

Demographics, diagnosis, engagement, barriers

02

Build Random Forest model

03

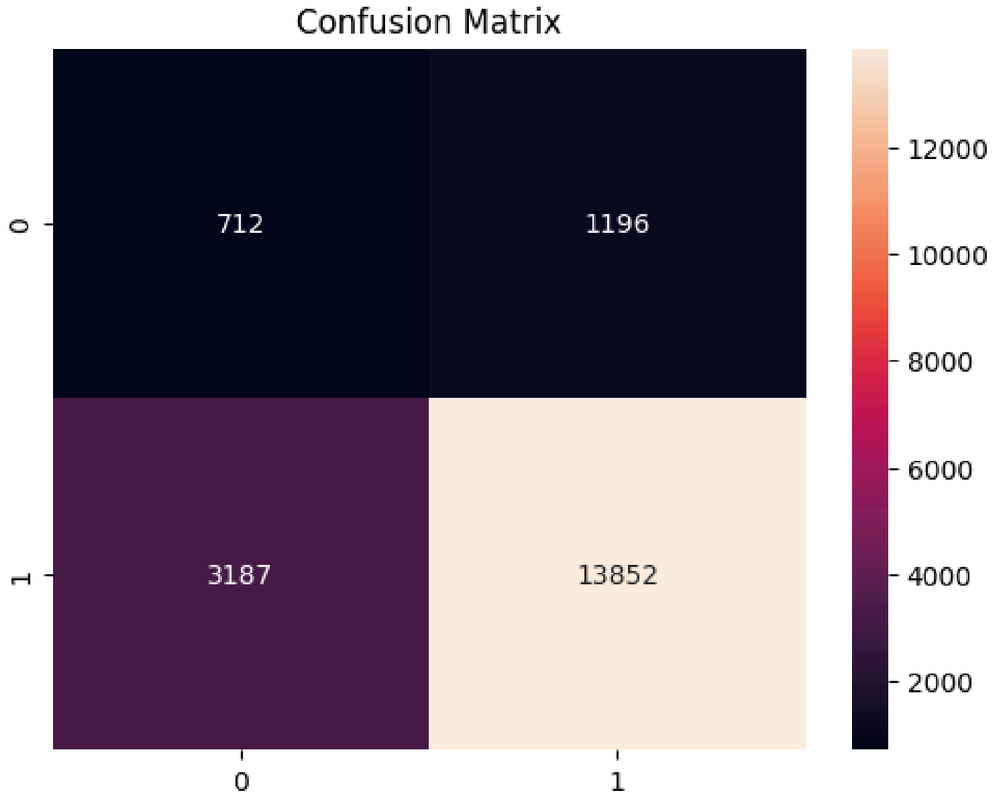
Predict most important factors of delay

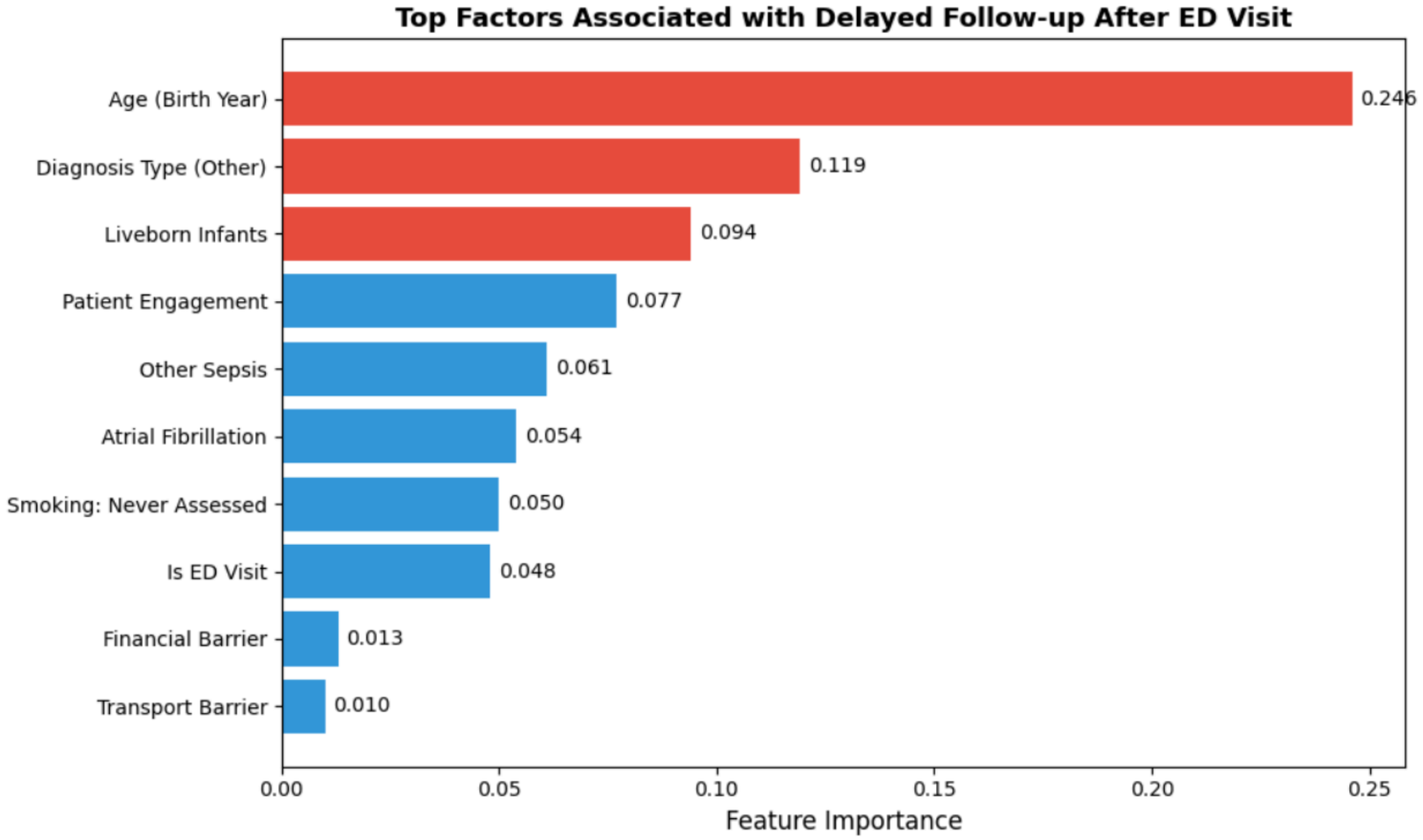
MODEL USED

- Model: Random Forest Classifier
- ROC AUC: 0.68 (moderate performance)
- Goal: Identify patients at risk of delayed follow-up

Confusion Matrix Accuracy: 77%

- Imbalanced dataset → more delayed than non-delayed patients
- Correctly identified many delayed patients
- Some false positives (overpredict delay)
- Fewer missed high-risk patients

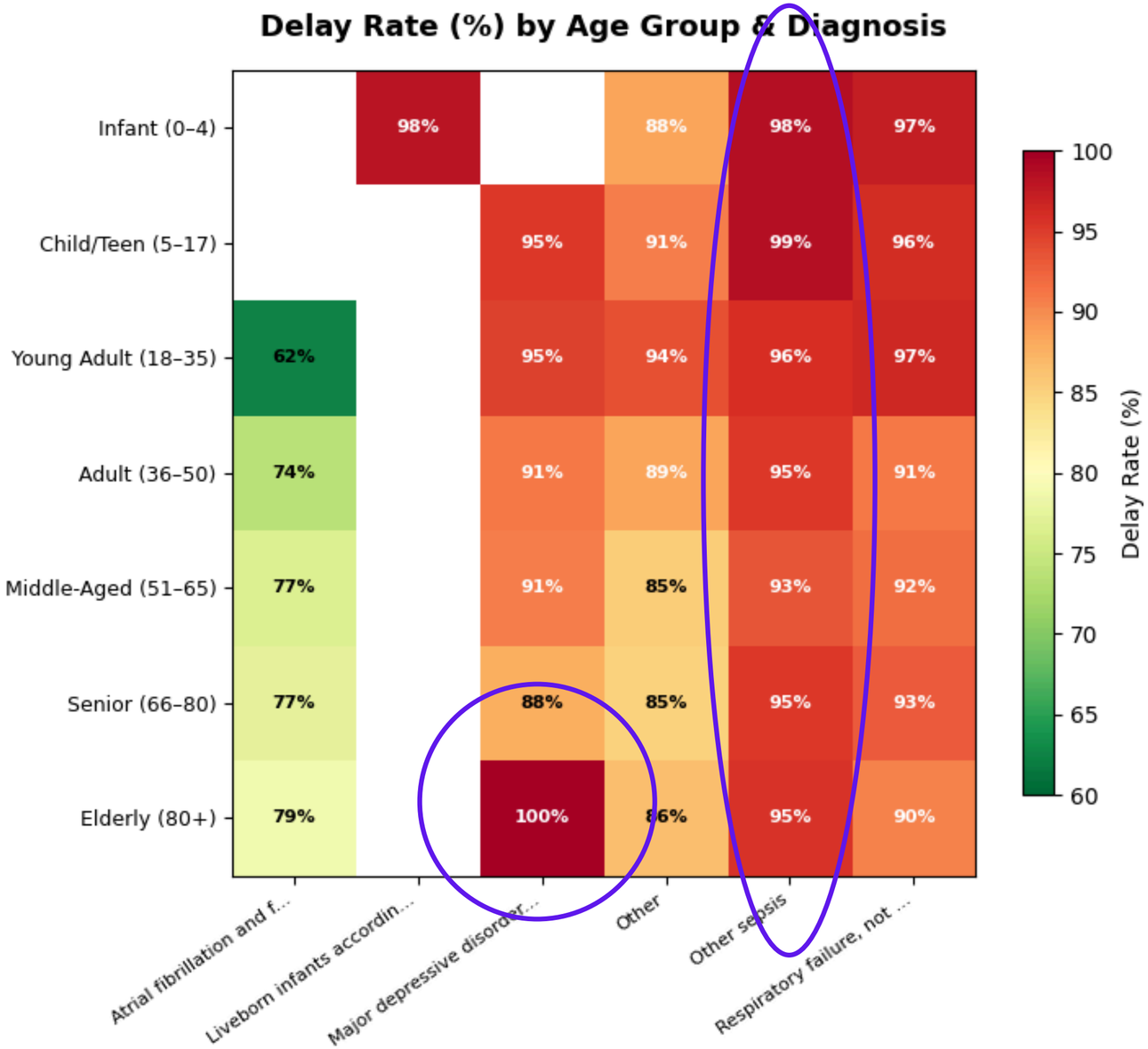




The Two Dominant Factors

- **Age** (0.246)
- **Diagnosis Type** (0.119)

We conducted a deeper analysis exploring how these two factors interact



- Most delay rates are **very high**, with many **above 90%**
- **Elderly patients with depressive disorder** show the **highest delay rate (100%)**
- **Other sepsis** has **consistently high delays** across age groups

These areas should be priority targets for intervention

RECOMMENDATION

- **Home-Based Follow-Up Care:** Implement a program where care workers conduct home visits after discharge, especially for elderly patients with mental health conditions.
- **Pre-book appointments at discharge:** Booking the next appointment before the patient leaves the ED
- **Age-Specific Outreach:** Follow-up behavior differs by age, so outreach should be applied

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THANK YOU

FOR YOUR ATTENTION